

Name: Siva Kumar M
Reg: 192521057
Subject: Cryptography and Network
Code: CSA-5117

Assignment - 1

1. Encrypt the message "Defend the wall" using Caesar cipher with key = 3

Ans: Plain text

Defend the wall

Shift each letter by +3:

D → G

E → H

F → I

E → H

N → Q

D → G

T → W

H → K

E → H

W → Z

A → D

L → O

L → O

Cipher text

GHI HQG GWKH ZDOO

94
Loop
90

(b) Decrypt "Khooz ZBuOoG" using Caesar cipher

Ans Shift each letter by -3:

K	H	O	O	Z	B	O	G
↓	↓	↓	↓	↓	↓	↓	↓
H	E	L	L	O	W	O	R

L D

Plain text:

HELLO WORLD.

(c) A Caesar cipher used to encrypt "secret" to VHFUHU
Determine the key used and decrypt WKLV Lu D WHVU

Ans. Compare the first letter

S(17) → V(21)

* Shift = +3

key = 3

Now decrypt:

WKLV Lu D WHVU

* W → T, K → H, L → I, V → S ⇒ This

* L → I, V → S → is

* D → A, W → T, H → E, V → S, W → T

Plain text -> This is a test.

⇒ Test.

Key ->

Using The Substitution Key

A → Q, B → W, C → E, D → R, E → T, F → Y, ... Z → M

encrypt the word "Five"

Q: F → Y, I → O, R → K, E → T

Cipher text = YOKT

* Decrypt "UJHHD QEB NRFZH YOLTK" using english frequency analysis

Ans: plain text:

"There are some words"

3. (a) encrypt "ATTACK AT palace" using keyword "Lemon"

Ans (i) Plain text:

ATTACK AT palace

(ii) Repeat the keyword

keyword: LEMON LEMON LEMON

plain text: ATTACK AT palace

3 Convert letter to number

Plaintext A T T A C K A T p a l a c e
 value 0 19 19 0 2 10 0 19 15 0 11 0 2 4

Key Word L E M O N L E M O N L E M O
 value 11 4 12 14 13 11 4 12 14 13 11 4 12 14

4 Add the value (mod 26)

Plain		0	19	19	0	2	10	0	19	15
Key		11	4	12	14	13	11	4	12	14
Sum		11	23	31	14	15	21	4	31	29
Mod 26		11	23	5	14	15	21	4	5	3
Cipher text		L	X	F	O	P	V	E	F	D

Plain		0	11	0	2	4
Key		13	11	4	12	14
Sum		13	22	4	14	21
Mod 26		13	22	4	14	17
Cipher text		N	W	E	O	S

Cipher text = L X F O P V E F D N W E O S

Decrypt "ONXEPKMAE GTU" using keyword "key"

i) cipher text: O N X E P K M A E G T U
 keyword: K E Y K E Y K E Y K E Y

ii) cipher : O N X E P K M A E G T U
 value : 16 13 23 4 15 10 12 0 4 6 19 20
 key : 10 4 24 10 4 24 10 4 24 10 4 24
~~sub~~
 sub : 6 9 -1 -6 11 -14 2 -4 -20 -4 15 -4
 mod 26 : 6 9 25 20 11 12 2 22 6 22 15 22
 plain: G J Z U L M C W G W P W //

Q4 Encrypt "HI" using hill cipher

Ans Given $H = I, I = S \Rightarrow P = \begin{bmatrix} 7 \\ 8 \end{bmatrix}$
 $\begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$

$$\begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} 7 \\ 8 \end{bmatrix} = \begin{bmatrix} 45 \\ 54 \end{bmatrix}$$

$$45 \div 26 = 19$$

$$54 \div 26 = 2$$

Cipher text = TC.

* A → MR, B → ETC, E → EFN, R → TEK

Cipher text → MRETCEFNTEHELMAU

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Q6

Decrypt "EANMTEFCUHEIREL" using Board

Ans

ABDOR

Plain text:
Meet the class

Q6

(a) Encrypt "hello" using One-Time pad key "XMCKL"

Ans

Plaintext	H	E	L	L	O
value	7	4	11	11	14
Key	X	M	C	K	L
value	23	12	2	10	11
Sum →	30	16	13	21	25
Mod 26 →	4	16	13	21	25

Cipher text → EONVZ

Q1.

Describe

Set-3
Assion...

Name:

Siva Kumar m

21057

117

Key:

Algorithm 3

1- Encrypt

Why is OTP Unbreakable under ideal conditions?

The key is completely random

The key length is equal to message

The key is only used once

Every possible plaintext is equally likely, so attackers cannot determine original message

Q7.

(a) Encrypt "We are Discoverd" using keyword "zebra"

Ans Keyword: z e b r a
w e a r e
d i s c o
v e r e d

Key word: A B E R Z

Original no: 5 3 2 4 1

A → EOD,

B → ASR,

E → EIE, R → RCE,

Z → WDV.

Cipher text → EODASREIE RCEWDV.

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(b)

Decrypt EVLNEACDTKSEAOROFODDEECUXUN
Key word: TRIAL

Ans

(i)

Arrange key word $\rightarrow$ AILRT

(ii)

Divide the Ciphertext into equal columns

(iii)

Rearrange the column back to original keyword (TRIAL)

(iv)

Read the matrix row by row

Plaintext: We are discovered flee at once

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(c)

Encrypt "Information Security" using keyword "hack"

Ans

(i) plaintext length = 19, keyword = 4, Row required = 5

(ii)

Add one padding character = x

$\rightarrow$ padded message $\rightarrow$

Information security x

(iii)

H A C K
I n f o
R M A T
I O N S
e c u r
i t y x

Cipher text $\rightarrow$ NMOCTFANUYIRIEOTSRX

Frequency analysis of monoalphabetic Substitution Cipher

Cipher text:

UZOSOVUOHXMOPUGPOZPEUSGWSZOPFPE SXUZU
ETUCC

Step 1:

Count the occurrence of letters

Cipher text

Frequency

Z	8
O	6
U	5
P	5
S	5
E	4
V	3
G	3

Step 2: Compare with English letter frequency
most common letters in English are

E, T, A, O, I, N

Probable mapping

Z → E,

O → T,

U → A, P → O, S → I

Step 3: partial decryption

A E ? I T A T ? ? T ? T ? O ? ? O T E O ?
I ? E ? ? I E T O ? O I ? ? A E
A ? ? ?

Step 4:

The cipher text is likely an English sentence related to information or security. With only frequency analysis, a partial decryption, is possible. the exact plaintext cannot be determined without additional substitution mappings.

frequency analysis suggest the mapping

Z $\rightarrow$ E

O $\rightarrow$ T

U $\rightarrow$ A

P $\rightarrow$ O

S $\rightarrow$ I

giving a partial English Plaintext.

Find the known Plaintext "This is the" with Ciphertext to identify substitution mappings and use these mappings to decrypt the ciphertext.

Ciphertext: B J W O J U Y M J U N X F X N F U O J U X
 Known Plaintext: This is the

Step 1:

Find the Substitution mappings.

Compare the Ciphertext with known Plaintext

Ciphertext	B	J	W	O	Y	M	N	X
Plaintext	T	H	S	I	E	A	M	

Substitution message

B → T

J → H

W → S

O → I

Y → T

M → E

N → A

X → M

Step 2: Decrypt the message remaining

Cipher	B J W	O Y M	N X	F	R	U
Plain	T H S	I T E	A M	A	L	E

Now replace each cipher text letter

Cipher text	BJJW	OJW	YUJW	NX	F	XNFGJW
Decryption	This	is	The	Am	A	Message

Plain text = This is a message.

Q10. Use Kasiski Examination and frequency analysis to estimate the key and decrypt the cipher text using Vigenere decryption $P = (C - K) \bmod 26$

Ans. Cipher text = ZICVTUONGRZGVTUAVZHCOYGLMGRJ

Step 1:

Use Kasiski Examination and frequency analysis
estimated key length = 3

Step 2:

Recover the key

Compare the letter frequency for each key position

Recovered key = key.

Set-3

ASSIGN

full pro

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Repeat The key

Plaintext:

Z	I	C	V	T	W	O	N	G	R	Z	G	V	T	W	A	V	Z	H	C
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

Key:

K	e	y	k	e	y	k	e	y	k	e	y	k	e	y	k	e	y	k	e
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

Cipher text:

O	Y	G	L	M	G	J
---	---	---	---	---	---	---

Key:

y	k	e	y	k	e	y
---	---	---	---	---	---	---

Step 4: Convert letters to numbers

Letter	Z	I	C	V	T	W	O	N	G
Value	25	8	2	21	19	22	16	13	6

Key

K	e	y	K	e	y	K	e	y
---	---	---	---	---	---	---	---	---

Value

10	4	24	10	4	24	10	4	24
----	---	----	----	---	----	----	---	----

Step 5: Decrypt using the formula

$$Z - K = 25 - 10 = 15 \Rightarrow P$$

$$I - E = 8 - 4 = 4 \Rightarrow E$$

$$C - Y = 2 - 24 = -22 + 26 = 4 \Rightarrow E$$

Step 6: Final plain text

Recovered key = key,

Plain text = We are discovered save yourself